



Cracking the Trending Code: How YouTubers Optimize Metadata to Go Viral

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Process Book

Final website: <https://vizify-seven.vercel.app>

Screencast: <https://youtu.be/u6vEdgHtyEo>

0.1 Introduction: the idea

YouTube visibility depends not only on video quality, but also on how a video is presented. Titles, tags, descriptions, categories and publishing time are all metadata choices that creators can control before uploading. Our project asks: *How do YouTubers optimize metadata to reach the trending page?*

Using a dataset of YouTube trending videos, Vizify turns this question into an interactive dashboard for creators and media students. The goal is not to build a perfect virality predictor, but to reveal recurring metadata patterns across countries and categories: when trending videos are published, how titles are written, which tags appear often, and how descriptions relate to engagement.

The motivation behind this angle is that the trending page often looks like a neutral ranking of popular videos, while creators know that visibility is also shaped by presentation strategies. A title can be written as a question, include numbers, use emojis, or rely on stronger promotional language. Tags can be broad and generic, or very specific to a niche. Descriptions can be minimal or detailed. By visualizing these choices, the project studies the practical layer between content creation and algorithmic visibility.

Our intended user is someone who wants to understand patterns rather than read a statistical report. For that reason, the dashboard is organized around questions a creator might ask before publishing: when should I post, what title style should I use, which tags seem useful, and how much effort should I put into the description? This framing guided the whole design process.

0.2 Initial concept and first sketches

During the first milestone, we explored several possible angles: country differences, category performance, views/likes correlation and metadata design. We narrowed the project to the metadata fields creators directly control: title, tags, description and pub-

lish time. This made the project more original than a general popularity dashboard, because it focuses on the design choices behind trending videos.

Our first exploration confirmed that the dataset was coherent enough to support this direction. Views and likes followed the expected positive relationship, but the plot also showed outliers and category differences. This encouraged us to move beyond global averages and design a tool where users could isolate countries, categories and metadata strategies.

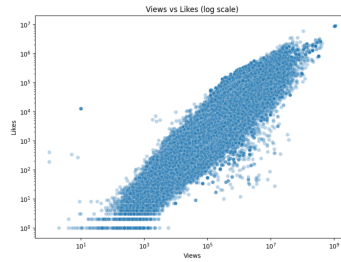


Figure 1: Early exploration confirmed a strong relation between views and likes, while also revealing outliers.

The first Figma sketches were already organized around five creator-facing questions. The final implementation kept the same analytical structure, but simplified the visual style and added clearer interactions. The sketches were useful because they forced us to decide early how each question would be represented visually instead of only listing possible analyses.

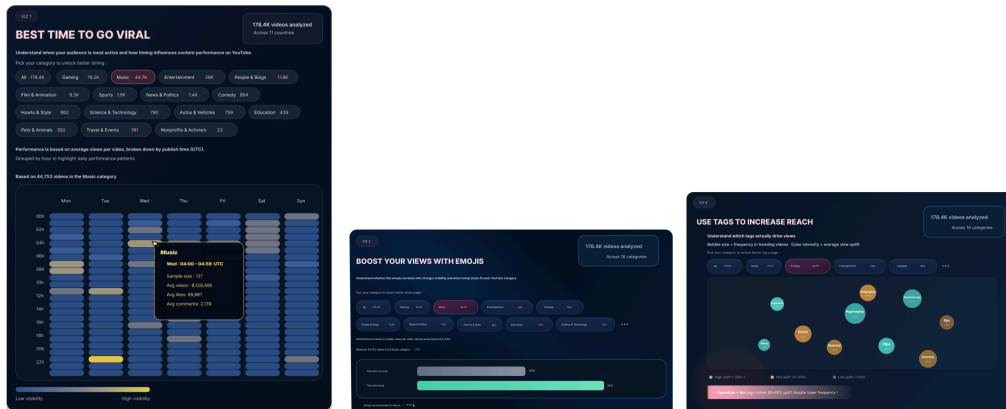


Figure 2: Initial sketches for the time, emoji and tag views.

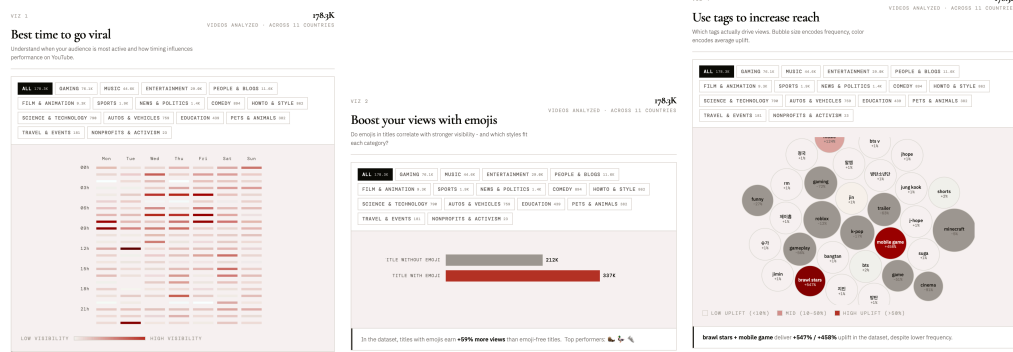


Figure 3: Final implemented versions of the same views.

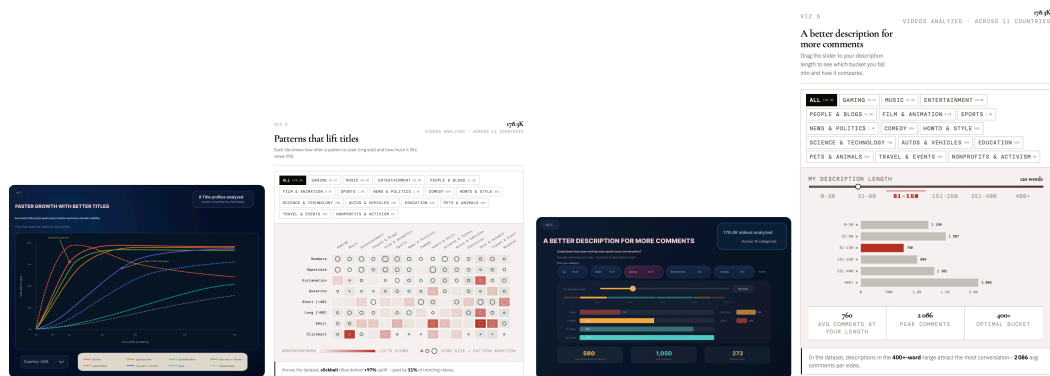


Figure 4: Title-pattern and description views evolved from simple sketches into more actionable final charts.

Time heatmap. The initial idea used day and hour as a grid, with color representing performance. This survived almost unchanged because it was the clearest way to show publishing windows. The final version added metric toggles and better labels.

Emoji and title patterns. We first planned simple comparisons of title styles. In the final version, the emoji chart stayed simple, while title patterns became a matrix where size shows adoption and color shows view uplift.

Tags and descriptions. Tags were first imagined as bubbles because a list would be too dense. Description length became a bucketed bar chart with a slider so users can compare their own description length with the dataset. These two views are the most directly actionable: the user can identify which keywords appear promising and benchmark their own description length against trending videos.

The sketches also helped us reject some ideas. We considered adding more global charts about category popularity and country ranking, but these did not directly answer the metadata question. Keeping them would have made the project feel broader but less focused. Instead, we used category and country as filters, so they support the story without becoming the story themselves.

0.3 Design iterations and evolution

The main evolution was moving from a collection of charts to a guided data story. We added an overview, key questions and a carousel dashboard so users can explore one visualization at a time. The color palette also changed from a dark theme to a lighter interface, which improved label readability and made the charts feel less crowded.

Each chart was refined during implementation. The heatmap gained a metric toggle between views, likes and comments. The emoji chart gained clearer labels and dynamic insights. The title-pattern view added sorting and a top-uplift mode. The tag chart added a switch between frequent tags and strongest uplift. The description chart added a slider and direct comparison with the best-performing bucket.

This iteration process also changed how we wrote the explanations. In the first sketches, some views were still described as general performance charts. In the final version, every chart is introduced by a user question. This made the interface more coherent: users do not need to know the dataset columns before interacting with the dashboard. They can start from a practical need, then use the chart to understand the pattern behind it.

Another important iteration was the move from static comparison to scoped exploration. In early drafts, most charts were imagined as global results over the whole dataset. During implementation, we realized that a global answer could be misleading because YouTube categories behave very differently. For example, an emoji may be normal in entertainment content but unusual in education or news. The shared country and category filters were therefore added as a central part of the experience.

We also refined the amount of text displayed in the interface. The first design drafts included more explanatory text around each chart. In the final version, we reduced long descriptions and relied on short titles, legends, tooltips and dynamic insight labels. This made the page lighter and allowed the charts to stay central.

0.4 Design decisions and challenges

Audience framing. The dashboard is designed for creators and communication students, so we framed every view as a practical question instead of a technical metric. This helped us avoid an overloaded dashboard and keep only variables that support the story: timing, emojis, title patterns, tags and descriptions.

Encoding choices. We selected simple encodings whenever possible: a heatmap for day/hour publishing patterns, bars for emoji comparison, a matrix for title patterns, bubbles for tags and buckets for description length. The goal was to make each chart readable without requiring long explanations.

Data preparation. The raw dataset contains repeated entries for videos that stayed trending across several days. To avoid counting the same video too many times, pre-

processing keeps one representative row per country and video, using the highest-view entry. Tags were cleaned by splitting, lowercasing and deduplicating values. Description length was grouped into buckets so that users could compare ranges instead of individual character counts. Title patterns were detected with transparent rules for numbers, uppercase, questions, exclamation marks, length, emojis and clickbait-like words.

Technical choices. We precomputed aggregates into static JSON files instead of calculating everything in the browser. This made the React/D3 dashboard faster and easier to deploy. A shared country/category filter controls all charts, even though each visualization uses a different data shape.

Readability challenge. Several early versions tried to show too many labels at once, especially in the tag chart and title-pattern matrix. We reduced this by using tooltips, sorting modes and shorter labels. This was an important design tradeoff: the visualizations needed to remain informative, but the first screen of each chart also had to be readable without zooming.

Balancing precision and clarity. Some metrics could have been computed in several ways. For example, tag performance could be shown as raw average views, median views, or relative uplift compared with the selected scope. We chose uplift for the final tag and title-pattern views because it better communicates whether a metadata choice stands above the local baseline. At the same time, we kept the wording careful, because uplift is still descriptive and not causal.

Interaction challenge. The carousel design made the site easier to navigate, but it also required consistency. If every slide had completely different controls, the dashboard would feel fragmented. We therefore reused similar interaction patterns: hover for detail, buttons for mode changes, and filters for scope. The description view is the only one with a slider, because its question is naturally about placing a user's own value inside a range.

Limitations. The results show associations, not causation. A tag or title pattern linked to higher views does not necessarily cause a video to trend. Channel size, thumbnails, recommendation dynamics and external promotion are not available in the dataset, so the dashboard should be read as an exploration of metadata patterns rather than a guaranteed recipe for success.

This limitation influenced the tone of the final interface. We avoided saying that a pattern *guarantees* more views. Instead, the dashboard uses language such as comparison, association and uplift. This keeps the visualization useful while remaining honest about what the data can actually prove.

0.5 Final result and user experience

The final result is an interactive web dashboard called *Vizify*. It starts with a short overview, then presents key questions that guide the user toward the relevant visualization. The main dashboard is a carousel with five modules, all controlled by shared country and category filters.

Publish time. A heatmap shows when trending videos are published by day and hour. Users can compare countries, categories and metrics to identify common publishing windows.

Emojis. A bar chart compares titles with and without emojis, while also showing the most common emojis for the selected scope. This makes the result easy to interpret for creators deciding whether emojis fit their niche.

Title patterns. A matrix compares title styles such as numbers, questions, upper-case, short titles, long titles, emojis and clickbait-like wording. Ring size shows how often a pattern is used, while color shows relative view uplift.

Tags. The tag bubble chart separates frequent tags from high-uplift tags. Bubble size represents frequency and color represents relative performance, helping users see the difference between generic popularity and strategic value.

Descriptions. Description length is grouped into buckets and compared through average comments. A slider lets users place their own description length in the chart and compare it with the best-performing range.

The main interaction pattern is consistent across the site: choose a scope, inspect the visual pattern, then use tooltips or secondary controls for details. We avoided making the user configure too many parameters before seeing results. This was important because our target audience should be able to understand the dashboard quickly during the screencast and during independent exploration.

The final dashboard also works as a narrative sequence. The user can begin with timing, which is the most general publishing decision, then move toward more specific metadata choices: title style, tags and description. This order mirrors the way a creator prepares a video upload. First they decide when to publish, then they write the title, choose tags and complete the description.

One design detail we kept from the early sketches is the idea of actionable feedback. Each chart tries to end with a direct comparison: the strongest time cell, the emoji difference, the top title pattern, the highest-uplift tags or the best description bucket. This does not replace interpretation, but it helps the user leave each view with one clear takeaway.

Overall, the interface turns a large dataset into a set of creator-facing insights. It does not claim to predict virality, but it helps users understand which metadata strategies often appear around trending videos. The final experience therefore answers the

initial research question in a practical way: it shows how creators present videos that reach trending, and how these presentation strategies vary across niches.

0.6 Team contribution and peer assessment

The project was divided between research framing, data exploration, design, implementation, report writing, final debugging and screencast preparation. The main decisions were discussed collectively, especially the shift from a broad trending-video dashboard to a creator-oriented metadata story.

Member	Main contributions	Share
Imane Raihane	Research framing, data interpretation, process-book writing, narrative and presentation preparation.	33%
Imane Lahrichi	Figma sketches, visual design, design iterations, report writing and visual refinement.	33%
Victor Zablocki	Screencast preparation, final presentation, interface testing and final debugging support.	34%

Table 1: *Approximate distribution of work.*

The split is approximate because several tasks overlapped. All members contributed to the final submission, but in different roles.

Collaboration was especially important during the final milestone because the process book, application and screencast had to tell the same story. We therefore aligned the report structure with the dashboard structure: idea, sketches, iterations, final result and contribution. This made the final submission more coherent and reduced the risk of presenting disconnected deliverables.

0.7 Conclusion

Vizify evolved from a general exploration of YouTube trending videos into a focused data story about metadata optimization. The final dashboard answers the initial question by showing how publish timing, emojis, title patterns, tags and description length appear in trending content across countries and categories.

The main lesson was that data visualization is not only about choosing charts, but also about deciding what to leave out. Preprocessing, deduplication, filtering and interaction design all shaped the final message. By keeping the interface centered on creator questions, the project became more accessible and more useful.

The project remains limited by the dataset: it cannot observe thumbnails, channel reputation or recommendation-system behavior, and it cannot prove causality. However, it provides a clear way to explore metadata patterns and understand how creators present videos that eventually reach the trending page.

If we had more time, the next step would be to compare metadata patterns with temporal trajectories: for example, whether some title styles help videos stay trending longer, not only reach a high number of views. Another extension would be to integrate thumbnail features or channel-level history, which would give a more complete picture of YouTube visibility. Even without these extensions, the final project demonstrates how a large platform dataset can be transformed into a focused and interactive visual explanation.